

Verify Numbers and Dates

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of claim support and citation evaluation.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multimodal Evidence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 16: Verify Numbers and Dates

Precise values require direct evidence and recalculation.



CREATOR MISSION

Cross-check dates and recompute a percentage claim.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Building a grounded answer. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can verify numbers and dates without inventing the intended correction.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use verify numbers and dates to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

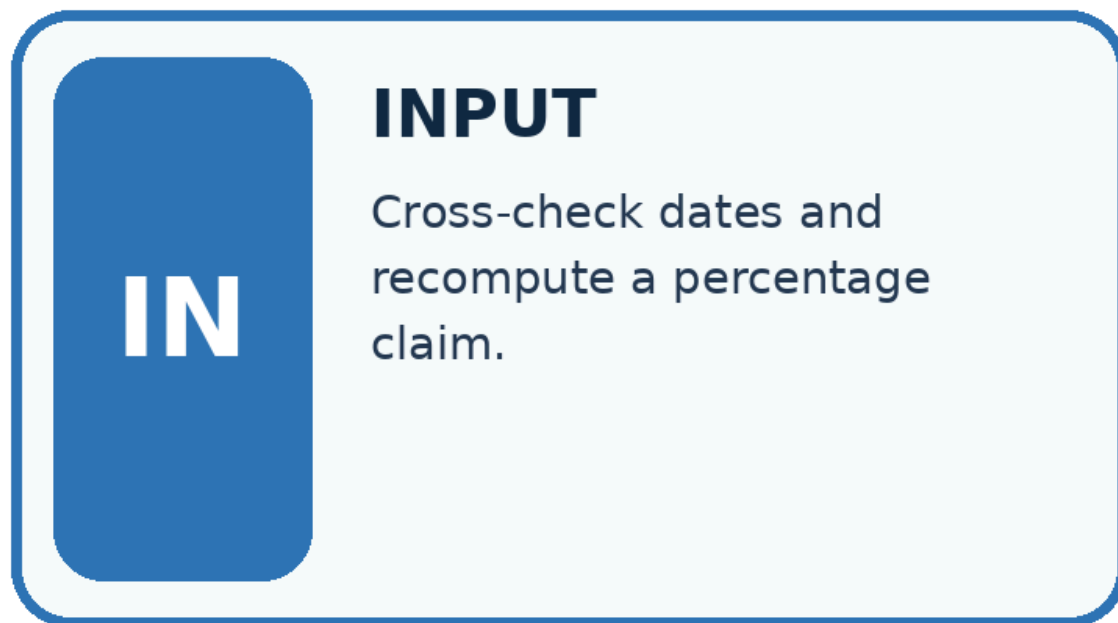
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Precise values require direct evidence and recalculation.





EVIDENCE

Cross-check dates and recompute a percentage claim.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

Claim support and citation evaluation — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

Check or calculation: For this classroom rubric count a claim as supported only when its cited passage entails it.

Conclusion and limit: Claim support is $2/4=50\%$; citation presence is $3/4=75\%$; link existence among three citations is $3/3$. These measure different failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The count is revised to 14 passes out of 20, but no date record is added.

Prediction: 70% now matches the count; June 31 remains invalid and unresolved.

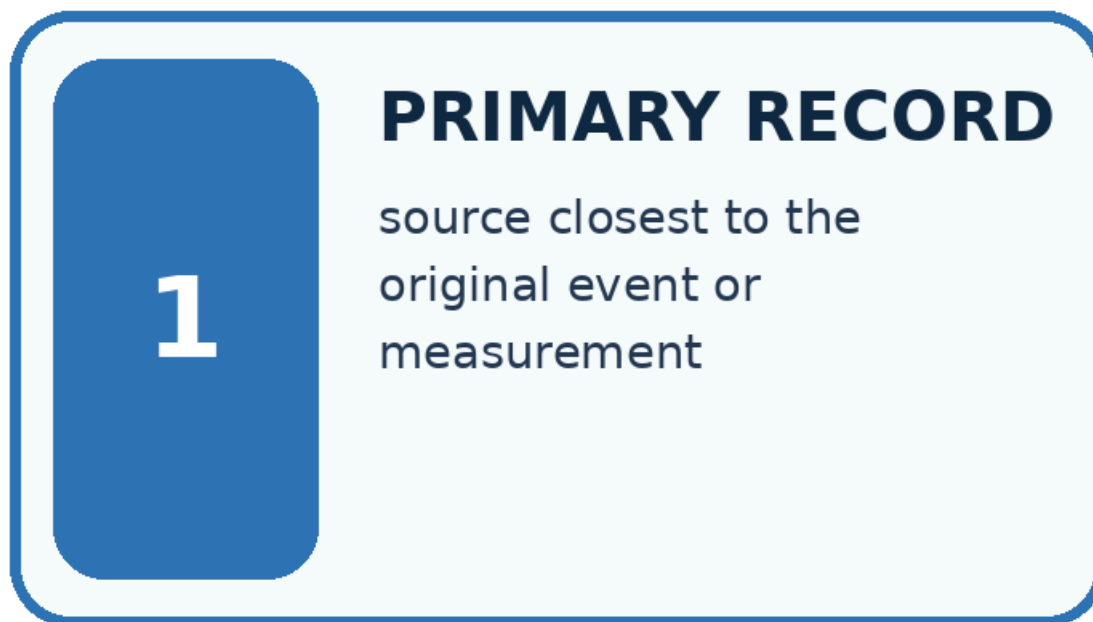
Reason: $14 \div 20 \times 100 = 70\%$; changing a count provides no new calendar evidence.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

RECALCULATE

independently repeat
arithmetic

3

UNIT

scale attached to a value

4

AS-OF DATE

time when information was current

Word · My own example · Not this / boundary

Word: primary record

My own example: An original record directly relevant to a claim, rather than a later summary of it.

Not this / boundary: _____

Word: recalculate

My own example: Compute a quantity again from the original values or corrected inputs.

Not this / boundary: _____

Word: unit

My own example: An agreed amount used to express a measurement, such as centimetres or seconds.

Not this / boundary: _____

Word: as-of date

My own example: The date up to which a claim or record is stated to be current.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Entailment

Meaning: Whether the cited passage supports the specific claim

Word: Citation coverage

Meaning: Which claims include source references

Word: Link validity

Meaning: Whether a source reference resolves to the intended record

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 16 • WORKED EVIDENCE

- 1 Claim: 18 of 24 tests passed, reported as 80%.
- 2 Recalculate: $18 \div 24 = 0.75 = 75\%$.
- 3 Check denominator, unit, rounding, and source date.
- 4 Correct the claim and preserve the calculation trail.

1. Claim: 18 of 24 tests passed, reported as 80%.
2. Recalculate: $18 \div 24 = 0.75 = 75\%$.
3. Check denominator, unit, rounding, and source date.
4. Correct the claim and preserve the calculation trail.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

Claim support is $2/4=50\%$; citation presence is $3/4=75\%$; link existence among three citations is $3/3$. These measure different failures.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Verify Numbers and Dates

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

The facts we will use

Record: 15 of 20 fictional model trials passed. Draft: "70% passed; the meeting is on June 31."

June has 30 days. No authoritative record gives the intended meeting date.

Verify Numbers and Dates

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check denominator

There are 15 passes out of 20 total trials.



2. Calculate percentage

$15 / 20 \times 100 = 75\%$, not 70%.



3. Validate date

June 31 is invalid because June has 30 days.



4. Resolve missing intent

Ask the event owner for the correct date; do not assume June 30 or July 1.

CHANGE ONE THING

The count is revised to 14 passes out of 20, but no date record is added.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Check denominator

What happens and why: There are 15 passes out of 20 total trials.

Step: 2. Calculate percentage

What happens and why: $15 \div 20 \times 100 = 75\%$, not 70%.

Step: 3. Validate date

What happens and why: June 31 is invalid because June has 30 days.

Step: 4. Resolve missing intent

What happens and why: Ask the event owner for the correct date; do not assume June 30 or July 1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended. These checks use different evidence: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

What this example does not establish

This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

Read the labelled picture: Which part can be corrected immediately from existing evidence, and which needs follow-up?

Change one thing: The count is revised to 14 passes out of 20, but no date record is added. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Specific numbers are reliable because they look precise. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Claim support and citation evaluation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which part can be corrected immediately from existing evidence, and which needs follow-up?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Audit six claims for source, date, unit, denominator, and arithmetic; mark verified, corrected, or unresolved.

Audit six claims for source, date, unit, denominator, and arithmetic; mark verified, corrected, or unresolved.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Route arithmetic to a calculator and dates to direct sources before presenting them.

THIS WEEK’S CHECKPOINT Route arithmetic to a calculator and dates to direct sources before presenting them.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Make Study Buddy check quantities, units, dates and source support separately; record unresolved fields instead of guessing.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Cross-check dates and recompute a percentage claim.

CHALLENGE Cross-check dates and recompute a percentage claim.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An answer says $15/20=80\%$ and sets a deadline of February 30.

The question

Verify both.

Worked explanation

$15/20=75\%$. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Verify a percentage independently

Evidence or stage · Value or result

Evidence or stage: Correct portion

Value or result: 15/20

Evidence or stage: Remaining portion

Value or result: 5/20

An answer says $15/20=80\%$ and sets a deadline of February 30.

Verify both.

$15/20=75\%$. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

14. Try a new case independently

Independent case: Changed independent case

A draft claims 18 of 24 learners is 60%, and gives an April 31 deadline. Check both statements without inventing a deadline.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A report says 12 of 40 responses is 40%. Recalculate and state the correction with the denominator.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Route arithmetic to a calculator and dates to direct sources before presenting them.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A report says $21/28=70\%$ and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of claim support and citation evaluation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of claim support and citation evaluation

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.